

Replication Instructions

Reproducing Results

Follow these steps to accurately replicate the results documented in the final report:

- 1. Install Dependencies: Install the required packages as outlined in the Requirements Document.
- 2. Directory Structure: Place the Datasets directory exactly one level above the codebase directory.
- 3. Run Script: Execute ``python evaluate.py`` in your terminal.

Determinism & Consistency

The evaluation involves 30 randomized iterations. By employing fixed random states for the `train_test_split` functions, the outputs for both the baseline tool (Naive Bayes) and the advanced tool (SVM) are completely deterministic up to floating-point precision on compatible setups.

Statistical Significance

The Wilcoxon signed-rank test is applied automatically during execution. The generated p-values and evaluation metrics will exactly match the numbers listed in the official project report.